

# NASSAU CANDY DISTRIBUTOR

## Product Line Profitability & Margin Performance Analysis

End-to-End Data Analytics Project Report

**WORKFLOW**

Excel → SQL → Power BI → Interactive One-Page Executive Dashboard

|         |               |              |              |               |
|---------|---------------|--------------|--------------|---------------|
| 10,194  | 8,549         | \$141,783.63 | \$93,442.80  | 65.91%        |
| Records | Unique Orders | Sales        | Gross Profit | Profit Margin |

*Prepared as a portfolio-ready data analytics case study*

**Prepared by: Sonu Kumar**

# 1. Executive Summary

## Project overview, business purpose and final outcome

This project analyzes sales, cost and profitability data for Nassau Candy Distributor and converts raw transactional records into an executive-friendly dashboard. The work follows a practical end-to-end analytics workflow: initial data inspection and validation in Excel, structured querying and aggregation in SQL, and interactive visualization in Power BI.

The primary objective is not simply to report revenue. The analysis focuses on profitability: which divisions and regions generate the most gross profit, which products contribute the most profit, and how profit margin differs across the leading products.

|                  |                 |                 |            |               |
|------------------|-----------------|-----------------|------------|---------------|
| <b>\$141.78K</b> | <b>\$48.34K</b> | <b>\$93.44K</b> | <b>39K</b> | <b>65.91%</b> |
| Total Sales      | Total Cost      | Gross Profit    | Units Sold | Margin        |

## Key conclusions

- Chocolate is the dominant profit-generating division, contributing \$88,824.62 in gross profit.
- Pacific is the highest gross-profit region at \$30,485.94, followed by Atlantic at \$26,973.70.
- The five leading products by gross profit are all Wonka Bar products, with the top product generating \$19,357.50.
- The overall gross-profit margin is 65.91%, indicating that gross profit represents roughly two-thirds of sales.
- The final Power BI page consolidates KPIs, division analysis, regional profitability, top-product performance, and slicers into one management-oriented view.

## Project deliverables

| Stage        | Primary output                       | Purpose                                                    |
|--------------|--------------------------------------|------------------------------------------------------------|
| Excel        | Cleaned/validated dataset and checks | Understand structure, types, blanks, duplicates and totals |
| SQL          | Reusable analytical queries          | Aggregate, rank and validate business metrics              |
| Power BI     | Measures + interactive dashboard     | Communicate insights and enable filtering                  |
| Final report | End-to-end documentation             | Explain methodology, findings and recommendations          |

## 2. Business Problem, Scope & Objectives

Turning transactional data into actionable profitability intelligence

### Business problem

A distributor can have strong sales while still underperforming on profitability if costs, product mix or regional performance are not understood. Management therefore needs a compact analytical view that answers where revenue is generated, where profit is generated, and which products deserve attention.

### Analytical questions

1. What are the overall sales, cost, gross profit, units sold and gross-profit margin?
2. Which product division contributes the most gross profit?
3. Which geographic region contributes the most gross profit?
4. Which products are the top five contributors to gross profit?
5. How do the leading products compare on total sales and profit margin?
6. Can management filter the dashboard by order date, region and division?

### Scope

- Transaction-level sales analysis using the supplied Nassau Candy Distributor dataset.
- Financial measures: Sales, Cost and Gross Profit.
- Operational measure: Units sold.
- Dimensions: Product, Division, Region, Customer/Order information and Order Date.
- Interactive reporting through Power BI with a one-page executive dashboard.

### Success criteria

| Criteria               | Target                                                               |
|------------------------|----------------------------------------------------------------------|
| Data quality           | Consistent fields, valid numeric values, correct date interpretation |
| Analytical correctness | Dashboard totals reconcile to source data                            |
| Business relevance     | Every major visual answers a clear business question                 |
| Usability              | Filters and KPIs are visible without navigating multiple pages       |
| Presentation           | Clean one-page layout suitable for portfolio/interview demonstration |

### Design principle

A dashboard should not be filled with visuals simply because canvas space is available. Each visual should earn its place by supporting a business decision.

### 3. Dataset Overview & Data Dictionary

#### Source structure and analytical fields

The source file contains 10,194 transaction records and 18 columns. The data covers orders placed from 02 Jan 2024 to 31 Dec 2025. The dataset contains transaction, customer, geography, product and financial attributes.

#### Field groups

| Group       | Fields                                                    | Analytical role                        |
|-------------|-----------------------------------------------------------|----------------------------------------|
| Transaction | Row ID, Order ID, Order Date, Ship Date, Ship Mode        | Order-level tracking and time analysis |
| Customer    | Customer ID                                               | Customer-level identification          |
| Geography   | Country/Region, City, State/Province, Postal Code, Region | Regional performance analysis          |
| Product     | Division, Product ID, Product Name                        | Product mix and profitability          |
| Financial   | Sales, Cost, Gross Profit                                 | Revenue, cost and profit analysis      |
| Operations  | Units                                                     | Volume analysis                        |

#### Core calculated business measures

| Measure            | Definition                 | Result       |
|--------------------|----------------------------|--------------|
| Total Orders       | Distinct count of Order ID | 8,549        |
| Total Sales        | Sum of Sales               | \$141,783.63 |
| Total Cost         | Sum of Cost                | \$48,340.83  |
| Total Gross Profit | Sum of Gross Profit        | \$93,442.80  |
| Total Units Sold   | Sum of Units               | 38,654       |
| Profit Margin %    | Gross Profit ÷ Sales       | 65.91%       |

#### Data quality checks

- Numeric fields were checked for appropriate numeric data types.
- Order Date was interpreted using day-first date parsing to avoid date ambiguity.
- Business totals were reconciled across the analytical workflow.
- Product, Division and Region fields were used as categorical dimensions.
- The final dashboard totals match the source-level calculations.

## 4. Excel Stage — Data Preparation & Validation

First layer of the analytics workflow

Excel was used as the first practical checkpoint before database analysis. The purpose of this stage was to understand the dataset, validate its structure, inspect the fields and establish trusted baseline totals.

### Excel workflow

7. Import the CSV data into Excel.
8. Inspect column names, data types and sample records.
9. Check dates, numeric measures and categorical fields.
10. Look for blanks, duplicates and obvious data-quality issues.
11. Create basic totals for Sales, Cost, Gross Profit and Units.
12. Validate that Gross Profit is consistent with the relationship Sales – Cost where applicable.
13. Use Excel results as a baseline for subsequent SQL and Power BI reconciliation.

### Useful Excel calculations

```
Total Sales      = SUM(Sales_Column)
Total Cost       = SUM(Cost_Column)
Total Gross Profit = SUM(GrossProfit_Column)
Total Units      = SUM(Units_Column)
Profit Margin     = Total Gross Profit / Total Sales
```

### Why Excel was included

- Excel provides fast visual inspection of raw data.
- It makes early validation easier before loading data into a database.
- It provides a simple benchmark against which SQL results can be checked.
- It demonstrates a common real-world workflow where analysts receive flat files before transforming them into structured reporting models.

### Validation baseline

Excel-level totals establish the baseline: Sales \$141,783.63, Cost \$48,340.83, Gross Profit \$93,442.80 and Units 38,654.

## 5. Excel Stage — Exploratory Analysis

### Moving from raw records to business questions

After validating the source, Excel can be used for quick exploratory summaries. PivotTables or equivalent aggregation techniques help identify the strongest divisions, regions and products before building the final SQL model.

### Recommended exploratory summaries

| Analysis               | Rows / dimension | Value / metric         | Purpose                               |
|------------------------|------------------|------------------------|---------------------------------------|
| Division profitability | Division         | Sum of Gross Profit    | Identify dominant product category    |
| Regional profitability | Region           | Sum of Gross Profit    | Identify strongest geographic markets |
| Product profitability  | Product Name     | Sum of Gross Profit    | Rank product contribution             |
| Product margin         | Product Name     | Gross Profit / Sales   | Identify profitable product mix       |
| Overall KPIs           | None             | Sales, Cost, GP, Units | Create management baseline            |

### Initial findings from the dataset

- Chocolate dominates gross profit with \$88,824.62.
- Other contributes \$4,333.45, while Sugar contributes \$284.73.
- Pacific leads regional gross profit at \$30,485.94.
- The leading product by gross profit is Wonka Bar -Scrumdiddlyumptious, at \$19,357.50.
- The strongest products are concentrated in the Chocolate division, creating a clear product-mix story for management.

### Transition to SQL

Once the business questions were clear, the next step was to move from spreadsheet-style exploration to repeatable SQL queries. SQL provides a more scalable and auditable way to aggregate, group, rank and validate the same metrics.

## 6. SQL Stage — Database Loading & Analytical Workflow

### Structured analysis and repeatable calculations

The SQL stage converts the flat-file dataset into a structured analytical environment. The database layer is useful for repeatability: the same business questions can be answered through queries instead of manually rebuilding spreadsheet summaries.

### SQL workflow

14. Create a database/schema for the project.
15. Create a table matching the source columns and appropriate data types.
16. Load the CSV records into the table.
17. Verify row counts and numeric totals.
18. Run grouped aggregations for division and region.
19. Rank products using aggregation and ordering.
20. Calculate profitability ratios and validate the results against Excel.
21. Use the validated outputs as the analytical foundation for Power BI.

### Illustrative table structure

```
CREATE TABLE candy_sales (
  row_id INT,
  order_id VARCHAR(50),
  order_date DATE,
  ship_date DATE,
  ship_mode VARCHAR(50),
  customer_id INT,
  country_region VARCHAR(100),
  city VARCHAR(100),
  state_province VARCHAR(100),
  postal_code VARCHAR(20),
  division VARCHAR(50),
  region VARCHAR(50),
  product_id VARCHAR(50),
  product_name VARCHAR(150),
  sales DECIMAL(12,2),
  units INT,
  gross_profit DECIMAL(12,2),
  cost DECIMAL(12,2)
);
```

### Why SQL adds value

- Repeatable calculations instead of manual spreadsheet operations.
- Efficient grouping and aggregation across thousands of records.
- Clear separation between raw data and analytical logic.
- Easy reconciliation of dashboard measures against source-level totals.
- Strong demonstration of practical SQL skills for a data analyst portfolio.

## 7. SQL Stage — Business Queries & Validation

From SQL aggregation to decision-ready metrics

### Overall KPI query

```
SELECT
    COUNT(DISTINCT order_id) AS total_orders,
    SUM(sales) AS total_sales,
    SUM(cost) AS total_cost,
    SUM(gross_profit) AS total_gross_profit,
    SUM(units) AS total_units
FROM candy_sales;
```

### Gross profit by division

```
SELECT
    division,
    SUM(gross_profit) AS gross_profit
FROM candy_sales
GROUP BY division
ORDER BY gross_profit DESC;
```

### Gross profit by region

```
SELECT
    region,
    SUM(gross_profit) AS gross_profit
FROM candy_sales
GROUP BY region
ORDER BY gross_profit DESC;
```

### Top products by gross profit

```
SELECT
    product_name,
    SUM(gross_profit) AS gross_profit
FROM candy_sales
GROUP BY product_name
ORDER BY gross_profit DESC
LIMIT 5;
```

### Validation results

| Metric             | Validated result |
|--------------------|------------------|
| Distinct Orders    | 8,549            |
| Total Sales        | \$141,783.63     |
| Total Cost         | \$48,340.83      |
| Total Gross Profit | \$93,442.80      |
| Total Units        | 38,654           |
| Profit Margin      | 65.91%           |

### Control check

The SQL results were used as a second independent validation layer before building the Power BI dashboard.

## 8. Power BI Stage — Data Model, Measures & Calculations

### Building the reporting layer

Power BI is the presentation and interactive analysis layer. The dataset is loaded into Power BI, business measures are defined, and visuals are connected to the measures and dimensions.

### Core measures used in the dashboard

```
Total Sales = SUM(Candy_Sales[Sales])

Total Cost = SUM(Candy_Sales[Cost])

Total Gross Profit = SUM(Candy_Sales[GrossProfit])

Total Units Sold = SUM(Candy_Sales[Units])

Total Orders = DISTINCTCOUNT(Candy_Sales[OrderID])

Profit Margin % =
DIVIDE([Total Gross Profit], [Total Sales], 0)
```

### Top-product ranking logic

```
Product Rank =
RANKX(
    ALL(Candy_Sales[ProductName]),
    [Total Gross Profit],
    ,
    DESC,
    DENSE
)
```

The ranking measure supports the final Top 5 Products – Sales & Margin table. It ranks products by gross profit while allowing the dashboard to display the associated sales and margin values.

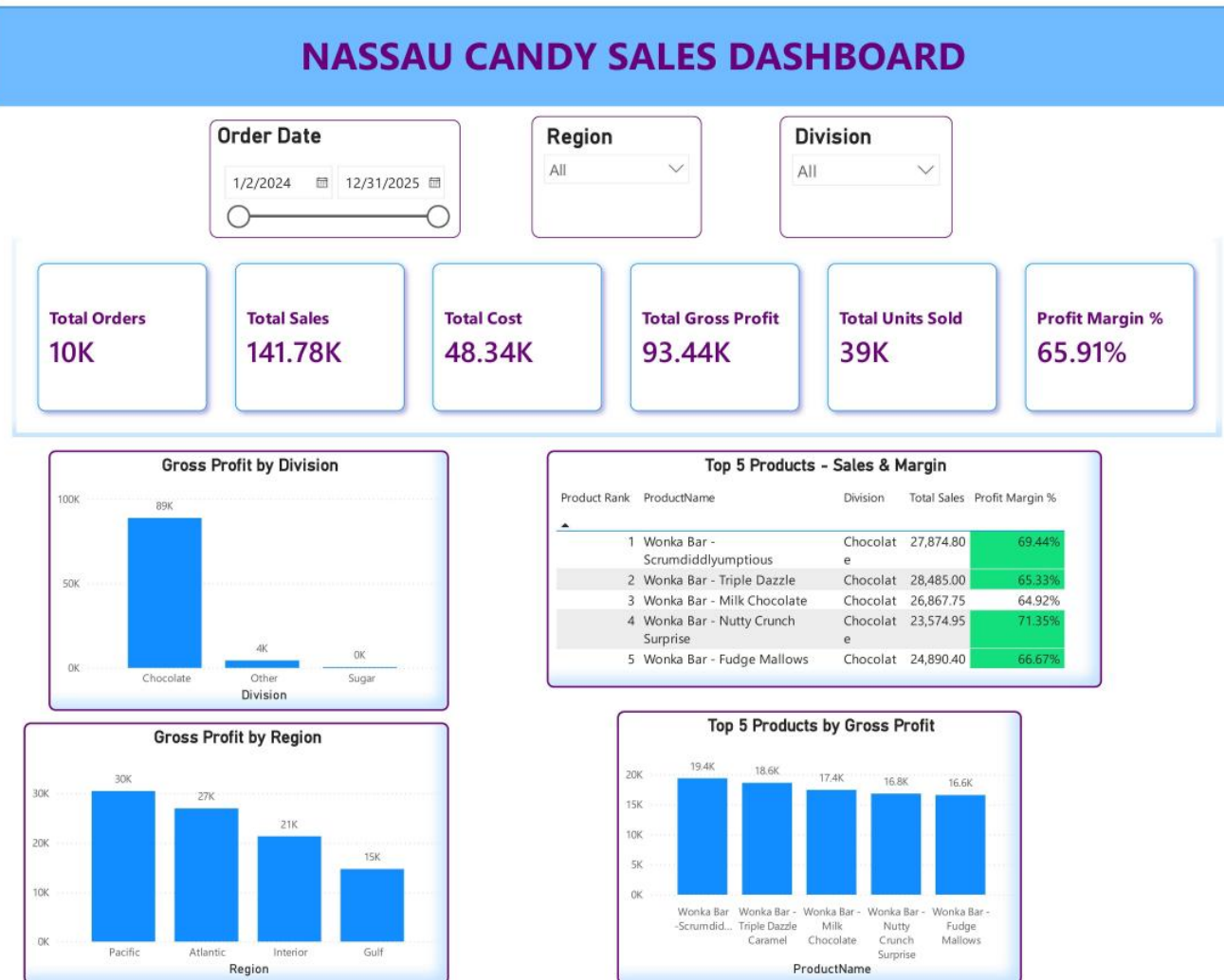
### Power BI reporting elements

| Element      | Purpose                                                                |
|--------------|------------------------------------------------------------------------|
| KPI Cards    | Fast view of total orders, sales, cost, gross profit, units and margin |
| Column chart | Gross profit by division                                               |
| Column chart | Gross profit by region                                                 |
| Top 5 chart  | Leading products by gross profit                                       |
| Table        | Top 5 products with sales and margin                                   |
| Slicers      | Order date, region and division filtering                              |

## 9. Dashboard Design & One-Page Executive View

A compact layout focused on decision-making

The final dashboard was deliberately redesigned as a one-page executive view. Instead of spreading important analysis across several pages, the dashboard places the most important KPIs and business visuals within one canvas.



Final one-page Power BI dashboard layout.

### Dashboard hierarchy

- Header: clear project title.
- Top controls: Order Date, Region and Division slicers.
- KPI strip: six headline measures.
- Analysis area: division profitability, regional profitability, top-product performance and sales/margin detail.
- Visual spacing: unused space was intentionally preserved rather than adding low-value charts.

## 10. Key Business Insights

What the analysis tells management

### 1. Chocolate is the primary profit engine

Chocolate contributes \$88,824.62 in gross profit, substantially higher than Other (\$4,333.45) and Sugar (\$284.73). This indicates that product mix is strongly concentrated around Chocolate.

### 2. Pacific is the strongest region by gross profit

Pacific generates \$30,485.94 in gross profit, ahead of Atlantic (\$26,973.70), Interior (\$21,282.49) and Gulf (\$14,700.67).

### 3. Top profitability is concentrated in Wonka Bar products

| Rank | Product                           | Gross Profit |
|------|-----------------------------------|--------------|
| 1    | Wonka Bar -Scrumdiddlyumptious    | \$19,357.50  |
| 2    | Wonka Bar - Triple Dazzle Caramel | \$18,610.20  |
| 3    | Wonka Bar - Milk Chocolate        | \$17,443.37  |
| 4    | Wonka Bar - Nutty Crunch Surprise | \$16,819.95  |
| 5    | Wonka Bar - Fudge Mallows         | \$16,593.60  |

### 4. Leading products have strong margins

| Product                           | Sales       | Profit Margin |
|-----------------------------------|-------------|---------------|
| Wonka Bar -Scrumdiddlyumptious    | \$27,874.80 | 69.44%        |
| Wonka Bar - Triple Dazzle Caramel | \$28,485.00 | 65.33%        |
| Wonka Bar - Milk Chocolate        | \$26,867.75 | 64.92%        |
| Wonka Bar - Nutty Crunch Surprise | \$23,574.95 | 71.35%        |
| Wonka Bar - Fudge Mallows         | \$24,890.40 | 66.67%        |

### 5. Overall profitability is healthy

Across the dataset, gross profit is \$93,442.80 on sales of \$141,783.63, resulting in a gross-profit margin of 65.91%. This KPI provides the high-level benchmark against which product and regional performance can be interpreted.

# 11. Recommendations & Business Actions

Turning analytical findings into next steps

## Product strategy

- Prioritize the leading Wonka Bar products for inventory availability and promotional planning.
- Use product-level margin alongside sales volume instead of relying on sales alone.
- Investigate lower-margin products separately to understand pricing or cost drivers.

## Regional strategy

- Study Pacific's successful product mix and commercial practices for possible replication in other regions.
- Review Gulf performance for opportunities to improve product mix, pricing or distribution efficiency.
- Use region slicers in Power BI to move from overall performance to market-specific analysis.

## Profitability management

- Track gross profit and margin together because high revenue does not automatically mean high profitability.
- Establish monthly or quarterly monitoring of sales, cost and gross profit.
- Create alerts or management thresholds for unusually low product margins.

## Dashboard usage

- Start with KPI cards to understand the current overall position.
- Use Region and Division filters to isolate performance.
- Use the Top 5 Products table to identify the products driving profitability.
- Use Order Date to compare periods when the reporting model is refreshed with new transactions.

### Recommended management question

Which products and regions should receive additional attention when sales growth and profitability move in different directions?

## 12. Challenges, Validation & Limitations

### Quality controls and practical considerations

#### Challenges addressed

- Date interpretation: the source dates required day-first parsing to avoid ambiguity.
- Measure consistency: totals were checked across the source dataset and analytical layers.
- Product ranking: a DAX ranking measure was required to create a reliable Top 5 view.
- Dashboard space: the report was redesigned to keep important analysis visible on one page without unnecessary charts.
- Visual hierarchy: KPI titles, values, filters and analytical visuals were arranged to make the dashboard readable at a glance.

#### Validation framework

| Layer          | Validation                                                    |
|----------------|---------------------------------------------------------------|
| Source / Excel | Row count, field inspection and baseline totals               |
| SQL            | Aggregations and grouped totals                               |
| Power BI       | Measures reconcile with validated totals                      |
| Dashboard      | Visuals respond to filters and retain consistent calculations |

#### Limitations

- The analysis is descriptive and does not forecast future sales or profit.
- No external pricing, competitor, promotional or inventory datasets were included.
- Gross profit is analyzed as supplied by the source; detailed cost-driver decomposition is outside the project scope.
- The report focuses on the most decision-relevant dimensions available in the dataset.

#### Future enhancements

- Add monthly sales and profit trend analysis.
- Add year-over-year or period-over-period comparisons.
- Introduce customer segmentation and customer profitability.
- Add product-level sales velocity and inventory metrics if inventory data becomes available.
- Automate data refresh through a production database or scheduled pipeline.

## 13. Conclusion & Project Skill Map

### End-to-end analytics capability demonstrated

The Nassau Candy Distributor project demonstrates a complete analytics workflow rather than a standalone dashboard exercise. The process begins with raw transactional data, moves through Excel validation and SQL analysis, and ends with a Power BI dashboard that communicates the most important business findings in a single view.

### End-to-end workflow

| Stage          | What was done                                        | Outcome                               |
|----------------|------------------------------------------------------|---------------------------------------|
| 1. Source data | Loaded and inspected CSV transactions                | Understood schema and business fields |
| 2. Excel       | Checked quality and calculated baseline KPIs         | Trusted validation benchmark          |
| 3. SQL         | Grouped, ranked and validated business metrics       | Repeatable analytical layer           |
| 4. Power BI    | Created DAX measures and visuals                     | Interactive reporting model           |
| 5. Dashboard   | Designed one-page executive view                     | Decision-ready communication          |
| 6. Insights    | Interpreted division, region and product performance | Business recommendations              |

### Skills demonstrated

- Excel: data inspection, cleaning/validation, formulas and exploratory analysis.
- SQL: SELECT, aggregation, GROUP BY, ORDER BY, DISTINCT COUNT and ranking-oriented analysis.
- Power BI: data loading, measures, DAX, ranking, cards, tables, charts and slicers.
- Data storytelling: KPI hierarchy, visual selection, dashboard layout and business interpretation.
- Business analysis: profitability, product mix, regional performance and margin analysis.
- Portfolio communication: documenting an end-to-end project from raw data to business insight.

### Final project takeaway

The strongest outcome of the project is the connection between technical analysis and business interpretation: the tools are not used independently; Excel establishes trust, SQL structures the analysis, and Power BI turns the validated analysis into an interactive decision-support product.

#### Portfolio positioning

This project can be presented as an end-to-end Data Analytics case study demonstrating Excel + SQL + Power BI workflow, business KPI development and dashboard storytelling.